

ARTIFICIAL INTELLIGENCE LAB

LAB 1 REPORT

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Program: **BS AI**

Semester: **3rd**

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Task 1: AI Milestone Timeline

The following timeline presents five important AI milestones, including an AI Winter period and modern compute/data milestones.

Year	AI Milestone	Importance
1950	Turing Test	Introduced an important way to discuss whether a machine can show human-like behavior through conversation.
1956	Dartmouth AI Workshop	Helped establish Artificial Intelligence as a formal research field.
1974–1980	First AI Winter	AI funding and interest decreased because many early expectations were not achieved.
2012	AlexNet + GPU Deep Learning	Showed the power of deep learning, large datasets, and GPUs for computer vision.

2017	Transformer	Introduced an important architecture that became a foundation for modern language AI.
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Sources

- Turing, A. M. (1950). Computing Machinery and Intelligence.
- Dartmouth College. The Founding and Future of AI.
- Stanford AI100. A Short History of AI.
- Krizhevsky, Sutskever & Hinton (2012). ImageNet Classification with Deep Convolutional Neural Networks.

Vaswani et al. (2017). Attention Is All You Nee

Task 2: Rule-Based Scholarship System

A rule-based system uses clear conditions to make a decision. A student is eligible only when all four conditions are satisfied.

Rules / Conditions

- Score must be 70 or above.
- All required documents must be complete.
- Attendance must be 75% or above.
- A recommendation letter must be available.

Final Rule: Score ≥ 70 AND Documents Complete AND Attendance $\geq 75\%$ AND Recommendation Letter Available $\rightarrow$ Eligible.

Note: The thresholds used in this lab are synthetic lab assumptions.

Four Core Test Cases

Test Case	Input	Expected	Actual	Reason
1	85, Complete, 80%, Yes	Eligible	Eligible	All conditions pass.
2	85, Complete, 70%, Yes	Not Eligible	Not Eligible	Attendance is below 75%.
3	85, Complete, 80%, No	Not Eligible	Not Eligible	Recommendation letter is missing.

4	70, Complete, 75%, Yes	Eligible	Eligible	Minimum thresholds are satisfied.
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Code and Output Evidence

The Python code was created and executed in VS Code. The file is submitted separately as lab1_rules.py. Paste the program output screenshot below.

Figure 1: Output of Rule-Based Scholarship System

```
Test Case 1
Expected: Eligible
Actual: Eligible
Result: PASS
-----
Test Case 2
Expected: Not Eligible
Actual: Not Eligible
Result: PASS
```

Task 3: Conversational Test vs General Intelligence

Comparison

A conversational test checks whether an AI can communicate in a human-like way. However, passing such a test does not prove general intelligence. An AI may give fluent and convincing answers but still make incorrect decisions or reasoning mistakes.

In the scholarship case, an AI may confidently say that a student is eligible even when the student's attendance is below the required 75%. Therefore, conversational ability should not be considered proof of reliable intelligence.

Additional Test: Scholarship Decision Accuracy Test

We give the system several cases with known correct answers and compare its actual results with the expected results. Boundary cases should also be included. This tests whether the system performs the actual task correctly and consistently.

Code and Output Evidence

The reliability test was implemented in lab1_compare.py and executed in VS Code. Paste the output screenshot below.

Figure 2: Output of Scholarship Decision Accuracy Test

```
Case 1
Expected: True
Actual: True
PASS
-----
Case 2
Expected: False
Actual: False
PASS
```

```
Case 3
Expected: True
Actual: True
PASS
-----
Case 4
Expected: False
Actual: False
PASS
-----
Case 5
```


Conclusion

This lab showed important AI milestones, a transparent rule-based decision system, and the difference between conversational ability and reliable task performance. The rule-based system uses clear conditions, while the accuracy test checks whether the system makes correct and consistent decisions.

References

- Turing, A. M. (1950). Computing Machinery and Intelligence.
 - Dartmouth College. The Founding and Future of AI.
 - Stanford AI100. A Short History of AI.
- Krizhevsky, Sutskever & Hinton (2012). ImageNet Classification with Deep Convolutional Neural Networks.
 - Vaswani et al. (2017). Attention Is All You Need.